Sampling Distributions and Unbiased Estimators

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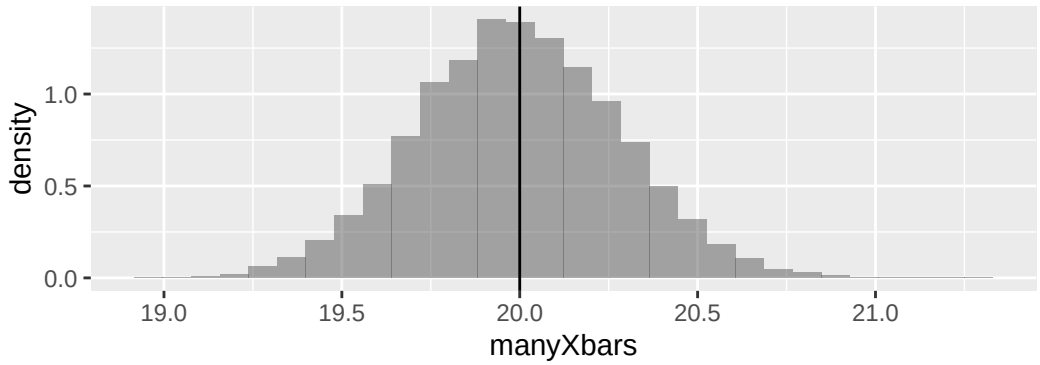
Method-of-Moment estimation

The program here will be comprised of sampling from a known distribution, using the Method of Moments to obtain an estimate $\hat{\theta}$ of parameter θ, and generating a simulated sampling distribution for $\hat{\theta}$.

Normal population, with $mu = 20, \sigma = 2$

First, we note that $\bar{X}$ is $\hat{\mu}$.

```
manyXbars = replicate(10000, {xvals = rnorm(50,20,2); mean(~xvals)})
gf_dhistogram(~manyXbars) |>
  gf_vline(xintercept=20)
```




```
mean(~manyXbars)   # note the nearness to 20
```

```
[1] 19.99538
```

This histogram gives the **sampling distribution of the sample mean** (or a simulation thereof), where

- iid random samples are drawn from Norm$(20, 2)$
- all samples are of size $n = 50$

The mean of the sampling distribution of $\bar{X}$—that is, $E(\bar{X})$—is computed as 19.9953792 which is very close to the mean $\mu = 20$ of the population, confirming that $\bar{X}$ is an unbiased estimator of the population mean μ.

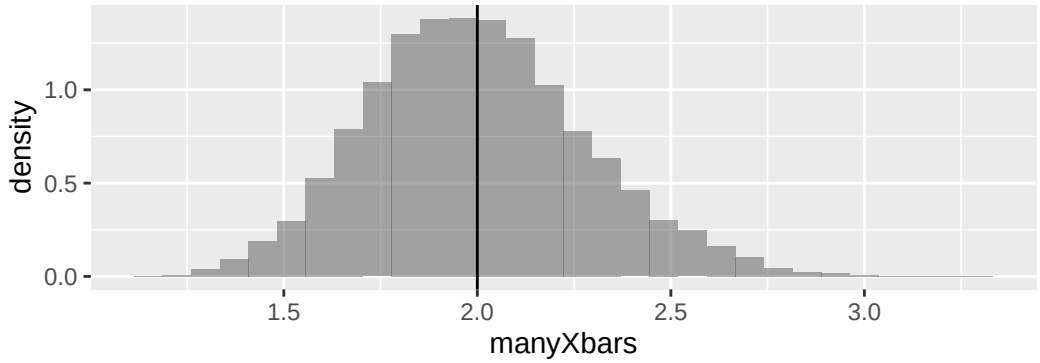
Q: Is this true merely because our population, in the last go, was *normal*? Let's try a non-normal population.

An exponential population with rate parameter

$\lambda = 0.5$

The mean $\mu = \frac{1}{\lambda} = 2$. We determined the method-of-moments estimator of λ is $\hat{\lambda} = \frac{1}{\bar{X}}$.

```
manyXbars = replicate(10000, mean(~rexp(50,0.5)))
gf_dhistogram(~manyXbars) |>
  gf_vline(xintercept=2)
```




```
mean(~manyXbars)   # note the nearness to 2
```

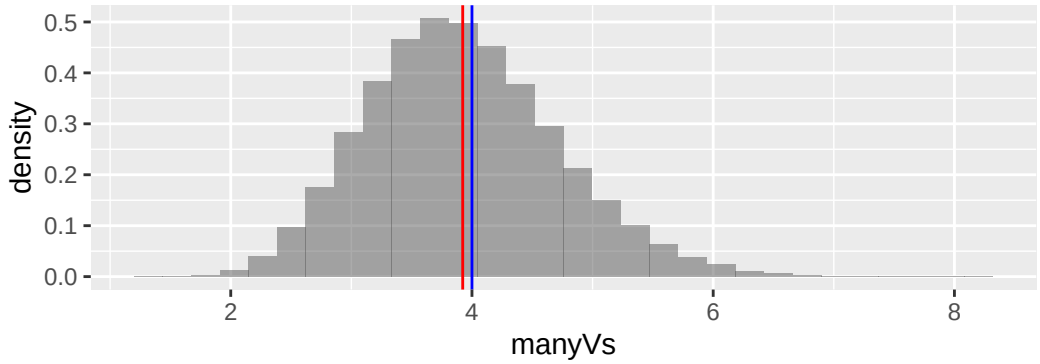
```
[1] 2.000193
```

Again, the sample mean appears to be an unbiased estimator of $\mu = \frac{1}{\lambda}$.

Exploring if $\hat{\mu}_2'$ is an unbiased of variance σ^2

Recall that $v = \frac{1}{n}\sum(X_i - \bar{X})$ is a the method-of-moment estimator of σ^2. Below, I return to sampling from a normal population.

```
manyVs = replicate(100000, 49/50 * var(~rnorm(50, 20, 2)))
gf_dhistogram(~manyVs) |>
  gf_vline(xintercept=4, color="blue") |>
  gf_vline(xintercept=mean(~manyVs), color="red")
```




```
mean(~manyVs)
```

```
[1] 3.923256
```

The value of the variance, $\sigma^2 = 4$, is highlighted with the blue vertical line. It is a bit to the right of the mean of the simulated sampling distribution of v (at the red vertical line), the value 3.923256, suggesting that v is *not* an unbiased estimator of σ^2. Verification of this can be done through direct calculation, but some details are technical. The result, however, is

$$\mathrm{E}(v) \;=\; \mathrm{E}\left[\frac{1}{n}\sum_i (X_i - \bar{X})^2\right] \;=\; \cdots \;=\; \frac{n-1}{n}\sigma^2,$$

not σ^2.